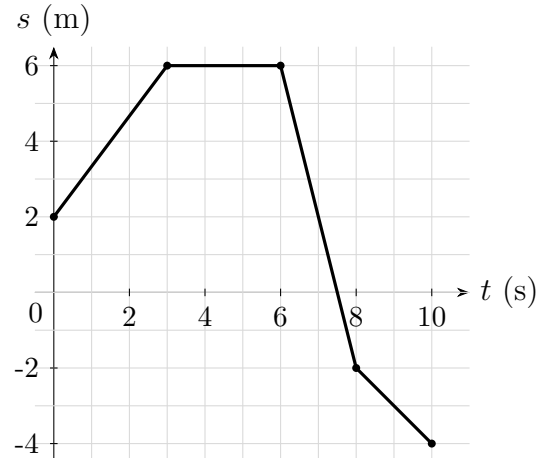


5.3 Classwork: Interpreting Motion Graphs — Solutions

1. The displacement s (metres) of a particle from the origin is shown for $0 \leq t \leq 10$.



- (a) Write down the initial displacement of the particle. [1]
- (b) Find the velocity during $0 \leq t \leq 3$. [2]
- (c) Describe the motion of the particle during $3 \leq t \leq 6$. [1]
- (d) Find the velocity during $6 \leq t \leq 8$. [2]
- (e) At what time does the particle pass through the origin? [2]
- (f) Find the total **distance** travelled from $t = 0$ to $t = 10$. [3]
- (g) Find the **displacement** from $t = 0$ to $t = 10$. [1]

Solutions

(a) $s(0) = \mathbf{2\text{ m}}$

(b) $v = \frac{6-2}{3-0} = \frac{4}{3} \approx \mathbf{1.33\text{ m s}^{-1}}$

(c) The particle is stationary (displacement is constant at 6 m).

(d) $v = \frac{-2-6}{8-6} = \frac{-8}{2} = \mathbf{-4\text{ m s}^{-1}}$

(e) The line from (6, 6) to (8, -2) has equation $s = 6 - 4(t - 6)$.

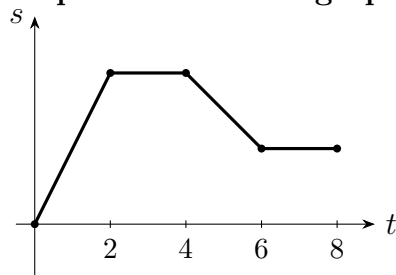
$$\text{Set } s = 0: \quad 6 - 4(t - 6) = 0 \Rightarrow t - 6 = \frac{6}{4} = 1.5 \Rightarrow t = \mathbf{7.5\text{ s}}$$

(f) Distance = $|6 - 2| + |6 - 6| + |-2 - 6| + |-4 - (-2)| = 4 + 0 + 8 + 2 = \mathbf{14\text{ m}}$

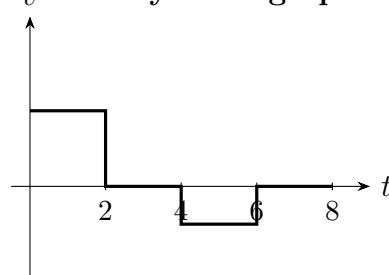
(g) Displacement = $s(10) - s(0) = -4 - 2 = \mathbf{-6\text{ m}}$

2. The graphs below show the displacement and velocity of a particle. Match each description (i)–(iv) to the correct time interval. [4]

Displacement–time graph



Velocity–time graph



- (i) The particle is moving forward at constant speed.
- (ii) The particle is stationary.
- (iii) The particle is moving backward at constant speed.
- (iv) The particle is stationary (for the second time).

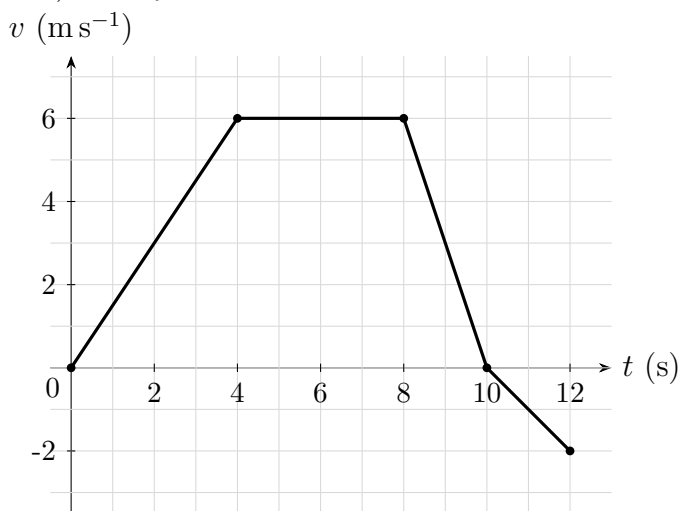
Solutions

Read both graphs together for each time interval:

- | | |
|--|--|
| $0 \leq t \leq 2$: s increasing, $v = 2$ (positive constant) | → (i) moving forward |
| $2 \leq t \leq 4$: s constant at 4, $v = 0$ | → (ii) stationary |
| $4 \leq t \leq 6$: s decreasing, $v = -1$ (negative constant) | → (iii) moving backward |
| $6 \leq t \leq 8$: s constant at 2, $v = 0$ | → (iv) stationary (second time) |

The key insight: positive v means s is increasing (forward motion); $v = 0$ means s is constant (stationary); negative v means s is decreasing (backward motion).

3. The velocity v (m s^{-1}) of a cyclist is shown for $0 \leq t \leq 12$ seconds.



- (a) Find the acceleration during $0 \leq t \leq 4$. [2]
- (b) Describe the motion during $4 \leq t \leq 8$. [1]
- (c) Find the acceleration during $8 \leq t \leq 10$. [2]
- (d) Find the total distance travelled from $t = 0$ to $t = 10$. [3]
- (e) What does the negative velocity during $10 < t \leq 12$ tell you? [1]
- (f) Find the total distance travelled from $t = 0$ to $t = 12$. [2]
- (g) Find the average speed of the cyclist over the 12-second interval. [2]

Solutions *Distance = area under the velocity–time graph.*

(a) $a = \frac{6 - 0}{4 - 0} = \frac{6}{4} = \mathbf{1.5 \text{ m s}^{-2}}$

(b) Constant velocity of 6 m s^{-1} (zero acceleration, moving forward at constant speed).

(c) $a = \frac{0 - 6}{10 - 8} = \frac{-6}{2} = \mathbf{-3 \text{ m s}^{-2}}$

(d) Triangle (0 to 4): $\frac{1}{2} \times 4 \times 6 = 12 \text{ m}$
Rectangle (4 to 8): $4 \times 6 = 24 \text{ m}$
Triangle (8 to 10): $\frac{1}{2} \times 2 \times 6 = 6 \text{ m}$
Total = $12 + 24 + 6 = \mathbf{42 \text{ m}}$

(e) The cyclist is moving in the opposite (negative) direction.

(f) Triangle (10 to 12, below axis): $\frac{1}{2} \times 2 \times 2 = 2 \text{ m}$. Total = $42 + 2 = \mathbf{44 \text{ m}}$

(g) Average speed = $\frac{44}{12} = \frac{11}{3} \approx \mathbf{3.67 \text{ m s}^{-1}}$